Agent Skill Discovery at Scale

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1 Category

Selected Category

Benchmark and Empirical Study

Topic

AI Agent Skill Discovery Strategies and Performance

2 One Sentence Claim

We claim that the relative accuracy and efficiency of agent skill discovery strategies depend on multiple factors, and we will show it by measuring final skill selection accuracy, query latency, and resource usage for each method against the baselines of LLM selection using the full catalog in a prompt and BM25 retrieval on progressively larger skill libraries constructed from the SkillRet dataset.

3 The Problem and Why It Matters

The use of skills allows AI agents to easily leverage additional capabilities and expertise on demand. Agent skills, as defined on agentskills.io, follow a standard that is well known in industry. Skills are loaded progressively by AI agents through three stages: discovery, activation, and execution. The first stage, discovery, is critical because this is where the AI agent selects the relevant skill. If this step is not performed correctly, it will affect the end result. This paper will focus on skill discovery, specifically the factors affecting performance and accuracy. The metrics considered are token usage, latency, and selection accuracy. The methods compared are prompt, BM25, vector, domain vector, and hybrid. Finally, this paper will evaluate these methods across different use cases: matches within a single domain or across multiple domains, overlapping skills, and cascading skills.

The description in a skill's header is limited to 1,024 characters. This limit makes it difficult to capture the full scope of a skill. As more organizations see the value of using, developing, and adding skills to AI agents, scalability issues may arise. For instance, an agent may fail to select a relevant skill, select the wrong skill, or select a less relevant skill. How can we improve the discovery step? How can we evaluate performance? How can we compare the performance of different skill discovery mechanisms?

4 Related Work

SkillRet provides the dataset and baselines for BM25 and dense retrieval used in our work, but evaluates retrieval on a fixed large skill pool rather than across progressively larger libraries [1].

Preprint.

Li (2026) shows that skill selection accuracy decreases as skill libraries grow and that hierarchical selection can help, while our work evaluates this scaling effect using real SkillRet skills and discovery through retrieval [2].

Graph of Skills evaluates libraries containing 200 to 2,000 skills and shows that bounded retrieval can reduce token growth, while our work compares simpler BM25, vector, and domain vector approaches without using a dependency graph [3].

Skill Shadowing shows that wrong skill selection, rather than context length alone, explains much of the degradation as more skills are added, motivating us to measure final LLM skill selection in addition to retrieval accuracy [4].

Skill Retrieval Augmentation (SRA) shows that successful retrieval does not guarantee that an LLM will correctly incorporate and use a skill, while our work focuses on how discovery strategy and library size affect the final skill selected by the LLM [5].

5 Approach

We will use the SkillRet dataset for benchmark testing. In addition to the skill dataset, test tasks are an important component of the benchmark. These tasks can be automatically generated or manually defined.

Data and Library Construction

The SkillRet dataset contains 6,006 agent skills collected from public repositories, along with 4,392 test queries and relevance labels (gold skills) that identify which skill or skills are relevant to each query [1]. The dataset is available under the Apache 2.0 license, which allows us to use it for this project free of charge. Each skill includes a name, a short description, and the full `SKILL.md` file.

We will define a test plan composed of individual test cases. Each test case consists of a task and the expected skill or skills to be selected. Using this test plan, we will run benchmarks across multiple scenarios by varying the following parameters:

- Number of skills: 10, 100, 1,000, 6,006
- Skill discovery method: prompt, BM25, vector, domain vector, hybrid
- Test task types: automatically generated or manually defined
- Use cases: tasks requiring one or more skills, overlapping skills, cascading skills
- Models: gpt-4o-mini, gpt-4.1-mini, gpt-4.1-nano, deepseek v4

Five Methods Are Compared

Prompt. All skill descriptions are included in the agent's initial prompt. The model selects the most relevant skills using its reasoning capabilities.

BM25. Ranks skills based on how well the words in the query match the words in each skill description.

Vector. Converts the query and skill descriptions into embeddings and ranks skills based on semantic similarity.

Domain vector. First selects the most relevant domain, then performs vector search only within that smaller group of skills.

Hybrid. Combines BM25 and vector retrieval.

Disclaimer

This proposal reflects our understanding at an early stage of the project. We expect to refine the scope and details as we learn more about this topic and conduct additional experiments. As described in the project's timeline, we plan to examine each skill discovery method in depth and review the selected research papers more closely. We will use an iterative process to characterize the skill discovery methods and review the testing strategies in order to obtain useful benchmark results.

6 Baseline and Trial Run Result

Following the two days of testing recommended for the project, we first downloaded and inspected the SkillRet dataset. The dataset is approximately 335 MB in total, so it can easily be processed on our available hardware. For this initial feasibility test, we evaluated 20 queries and retrieved the top five skills for each query. The retrieved skills were compared with the gold skills provided by SkillRet using Recall@5.

The BM25 baseline achieved a Recall@5 of **0.542** (54.2%) on this small trial of 20 queries. This result is not intended to support our final claim because the sample is too small. Its purpose was to confirm that the dataset can be downloaded and processed, that the gold skills can be used for evaluation, and that the retrieval pipeline and metric work correctly.

7 Evaluation Plan

We will compare the discovery methods across library sizes, LLMs, and embedding models using the following metrics:

- **Final selection accuracy:** Percentage of queries for which the correct skill or skills are selected.
- **Recall@5:** Percentage of relevant skills found in the top five candidates.
- **MRR@5:** Mean reciprocal rank of the first relevant skill among the top five candidates.
- **Token usage:** Average LLM input and output tokens per query. Embedding tokens are reported separately.
- **Relative latency:** Latency relative to a baseline measured on the same machine, allowing fairer comparisons across hardware.

Within each model configuration, all methods will use the same queries and skill library sizes. Hardware and model configurations will be recorded for reproducibility.

8 Compute Budget

Small models will be evaluated locally using the full test set of 4,392 queries. We estimate approximately 10 GPU hours for these experiments.

Medium and large LLMs will be evaluated through hosted APIs from providers such as OpenAI and OpenRouter using a representative subset of about 2,000 queries.

For DeepSeek V4 Flash 0731, assuming 300,000 input tokens per query and an input rate of $0.0215 USD per million tokens:

$$2{,}000 \times 300{,}000 = 600{,}000{,}000 \text{ input tokens.}$$

$$600 \times \$0.0215 \text{ USD} = \$12.90 \text{ USD} \approx \$18.40 \text{ CAD.}$$

At an assumed rate of $0.01 USD per million tokens for BGE-M3, embedding costs are expected to remain negligible compared with LLM costs.

Based on these estimates, the hosted benchmark budget is approximately $20 CAD per LLM configuration.

9 Timeline and Division of Work

Table 1 shows a tentative breakdown of the work planned for this project.

Weeks	Deliverable	Proposed lead
4	Project proposal	All team members
5	Clarify each skill discovery method; create a code sample; document each method	and
5	Review the selected academic papers in detail; summarize each paper; identify how this project adds value	
6	Validate the metrics; create the testing framework	and
6	Create the report skeleton; include the information clarified up to this point	
7	Execute benchmark testing; compile statistics; review results; adjust the testing framework	and
7	Review preliminary test results; compare with the reference academic papers; start documenting results	
8	Expand benchmark testing using different hosted models	
8	Expand benchmark testing using different local models	
8	Create the presentation skeleton; include key elements	
9	Consolidate results; write the report	All team members
10	Finalize the presentation	All team members
11	Presentation rehearsal; final report review; buffer time	All team members
12 to 13	Presentation, report, and reproduction review	All team members

Table 1: Proposed responsibilities and timeline.

10 Risks and Backup

The cost associated with using hosted LLMs is a risk. The backup plan is to use inexpensive hosted LLMs or free local LLMs. This may affect reasoning capabilities, but we can still structure the benchmark testing methodology, analyze previous research, and identify gaps, challenges, and opportunities.

References

[1] Ryangkyung Kang, Hongcheol Cho, and Youngeun Kim. SkillRet: A large-scale benchmark for skill retrieval in LLM agents. *arXiv preprint arXiv:2605.05726*, 2026. doi: 10.48550/arXiv.2605.05726.

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[3] Dawei Liu, Zongxia Li, Hongyang Du, Xiyang Wu, Shihang Gui, Yongbei Kuang, and Lichao Sun. Graph-of-Skills: Dependency-aware structural retrieval for massive agent skills. *arXiv preprint arXiv:2604.05333*, 2026. doi: 10.48550/arXiv.2604.05333.

[4] Hongwen Song and Song Wei. More skills, worse agents? skill shadowing degrades performance when expanding skill libraries. *arXiv preprint arXiv:2605.24050*, 2026. doi: 10.48550/arXiv.2605.24050.

[5] Weihang Su, Jianming Long, Qingyao Ai, Yichen Tang, Changyue Wang, Yiteng Tu, and Yiqun Liu. Skill retrieval augmentation for agentic AI. *arXiv preprint arXiv:2604.24594*, 2026. URL `https://arxiv.org/abs/2604.24594`.